

Transfer Learning for Domain Adaptation: Few-Shot Classification, Neural Style Transfer, and Domain Shift Adaptation from Clipart to Sketch

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Abstract—This paper presents a comprehensive transfer learning pipeline for domain adaptation between clipart and sketch representations using few-shot classification, neural style transfer, and domain shift adaptation. We address the challenge of transferring knowledge from a source domain (Clipart) to a visually distinct target domain (Sketch) across six clothing categories: jacket, pants, shoe, sock, t-shirt, and umbrella. Our methodology implements two transfer learning strategies (feature extraction and fine-tuning) using a pretrained ResNet-50 backbone, followed by Gatys-style neural style transfer to generate synthetic sketch-styled images from clipart content. Three domain adaptation strategies are compared: baseline without adaptation, target fine-tuning with 50 labelled examples per class, and style-transfer augmentation using synthetically generated images. Experimental results demonstrate that fine-tuning achieves 89.74% source accuracy but suffers a domain shift penalty of 27.84%. Target fine-tuning reduces this penalty to 4.80%, significantly outperforming both baseline and style augmentation approaches. Grad-CAM visualizations reveal that the unadapted model focuses on colour and texture cues absent in sketches, while domain-adapted models learn edge-based representations. Our findings confirm that direct fine-tuning on target domain data, even with limited labels, is the most effective adaptation strategy for the Clipart→Sketch domain shift.

I. INTRODUCTION

Transfer learning has emerged as a fundamental technique in computer vision, enabling models trained on large-scale datasets to be adapted to new tasks with limited labelled data [1], [2]. The ability to transfer knowledge across domains—from real photographs to cartoons, from medical scanners to another, or from simulators to real-world environments—is increasingly critical in practical applications [3]. However, domain shift remains a significant challenge: models that perform well on a source domain often degrade substantially when tested on a different target domain with similar semantics but different visual characteristics.

This paper addresses the domain adaptation problem between **Clipart** (source) and **Sketch** (target) domains using the DomainNet benchmark [3]. Our group (Group 6) focuses on six clothing categories: jacket, pants, shoe, sock, t-shirt, and umbrella. The choice of domains is particularly challenging

because sketch representations eliminate colour and texture entirely, retaining only edge-based contours—a stark contrast to the flat-colour, shaded clipart images.

The challenge is structured into three interconnected parts. **Part A** establishes a baseline few-shot classifier using a pretrained ResNet-50 [4] backbone, comparing feature extraction (frozen backbone) against fine-tuning (unfrozen higher layers). **Part B** implements Gatys-style neural style transfer [5] to generate synthetic training images that combine clipart content with sketch style, producing a form of unsupervised data augmentation. **Part C** evaluates three domain adaptation strategies: baseline without adaptation, target fine-tuning with 50 labelled examples per class, and style-transfer augmentation using the synthetically generated images.

Our contributions are threefold: (1) a systematic comparison of feature extraction and fine-tuning strategies under few-shot conditions (50 images per class); (2) a complete neural style transfer pipeline for generating sketch-styled images from clipart content; (3) an empirical evaluation of three domain adaptation strategies, quantifying the domain shift penalty and identifying the most effective approach.

The remainder of this paper is organized as follows. Section II reviews related work in transfer learning, neural style transfer, and domain adaptation. Section III describes our methodology, including data preprocessing, model architectures, and training protocols. Section IV presents experimental results, including quantitative comparisons and qualitative visualizations. Section V discusses our findings, limitations, and future directions. Section VI concludes the paper.

II. RELATED WORK

A. Transfer Learning and Few-Shot Classification

Transfer learning has become the de facto standard for training deep neural networks with limited labelled data [1], [2]. The core insight is that convolutional neural networks trained on large-scale datasets like ImageNet learn hierarchical feature representations, where early layers detect universal patterns (edges, textures, corners) and deeper layers become

increasingly task-specific [4]. Two primary strategies have emerged: *feature extraction*, where the pretrained backbone is frozen and only the classification head is trained; and *fine-tuning*, where deeper layers are unfrozen and trained with a low learning rate [1]. For few-shot scenarios—typically defined as training with 1-50 examples per class—feature extraction often provides a strong baseline, while fine-tuning can adapt higher-level features to the target distribution.

B. Neural Style Transfer

Gatys et al. [5] introduced neural style transfer, demonstrating that the content and style of images can be separated using the internal representations of a pretrained VGG-19 network [6]. Content is encoded in deep feature maps (typically `relu4_2`), while style is represented as Gram matrices of shallower feature maps, capturing pairwise feature correlations across spatial locations. The optimization process minimizes a weighted combination of content and style losses:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{content}} + \beta \mathcal{L}_{\text{style}} \quad (1)$$

where $\mathcal{L}_{\text{content}} = \frac{1}{2} \sum_{l \in \mathcal{C}} \|F^l - P^l\|_F^2$ and $\mathcal{L}_{\text{style}} = \sum_{l \in \mathcal{S}} \|G^l - \hat{G}^l\|_F^2$. For domain adaptation, style transfer provides a mechanism to generate synthetic target-domain images from source content, effectively performing unsupervised data augmentation [3].

C. Domain Adaptation and Domain Shift

Domain adaptation addresses the problem of distribution shift between source and target domains [2], [3]. The domain shift penalty is formally defined as:

$$\Delta_{\text{shift}} = \text{Acc}_{\text{source}} - \text{Acc}_{\text{target}} \quad (2)$$

Common adaptation strategies include: (1) *no adaptation*, serving as a lower bound; (2) *target fine-tuning*, where a small labelled target set is used to adapt the model; (3) *style-transfer augmentation*, where synthetic images bridge the domain gap; and (4) adversarial approaches such as Domain-Adversarial Neural Networks (DANN) [8], which train a domain classifier with a gradient reversal layer to learn domain-invariant features.

III. METHODOLOGY

A. Dataset Description

We use the DomainNet benchmark [3], a large-scale domain adaptation dataset containing six visual domains. Our Group 6 assignment is:

- **Source Domain:** Clipart (flat colours, simplified contours)
- **Target Domain:** Sketch (edge-only representations, monochromatic)
- **Categories** (6 classes): jacket, pants, shoe, sock, t-shirt, umbrella

The source dataset contains 617 labelled images, and the target dataset contains 1,756 labelled images. For few-shot training, we use 50 images per class (300 total) from the source domain.

B. Preprocessing Pipeline

To ensure fair comparison and avoid artefactual domain shift, identical preprocessing is applied to both source and target domains:

- Resizing to 224×224 pixels
- Standard ImageNet normalization: mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]
- Training augmentations: random horizontal flip (p=0.5), colour jitter (brightness=0.2, contrast=0.2, saturation=0.2, hue=0.1)

C. Part A: Few-Shot Classification

We employ a pretrained ResNet-50 [4] backbone with ImageNet-1K weights (IMAGENET1K_V2). Two strategies are compared:

1) *Feature Extraction (Frozen Backbone)*: All convolutional layers are frozen; only the final fully connected layer is replaced and trained:

$$\text{model.fc} = \text{Linear}(\text{in_features} = 2048, \text{out_features} = 6) \quad (3)$$

Training uses Adam optimizer with learning rate 1×10^{-3} for 30 epochs.

2) *Fine-Tuning*: After feature extraction training, layers 3 and 4 (the last two residual blocks) are unfrozen. Training continues with learning rate 1×10^{-4} for 30 additional epochs, using the pre-trained head weights as initialization.

Both strategies use batch size 32, cross-entropy loss, and Adam optimization. Train/validation/test splits are created with 50/10/remaining images per class.

D. Part B: Neural Style Transfer

We implement Gatys-style neural style transfer using a pretrained VGG-19 network [5], [6]. The architecture extracts:

- **Content layer:** `relu4_2` (deep feature map capturing object structure)
- **Style layers:** `relu1_1`, `relu2_1`, `relu3_1`, `relu4_1`, `relu5_1` (multi-scale texture representations)

The Gram matrix at layer l is computed as:

$$G_{ij}^l = \frac{1}{C_l H_l W_l} \sum_k F_{ik}^l F_{jk}^l \quad (4)$$

The total loss is optimized using the L-BFGS optimizer with parameters $\alpha = 1.0$ (content weight) and $\beta = 1 \times 10^4$ (style weight) for 300 steps. One style image per class is selected from the target domain, and 30 stylized images per class (180 total) are generated. These synthetic images combine clipart content with sketch style, serving as unsupervised data augmentation for Part C.

E. Part C: Domain Adaptation Strategies

Three adaptation strategies are evaluated:

1) *Strategy 1: Baseline (No Adaptation)*: The fine-tuned model from Part A is evaluated directly on the target domain test set. This establishes the lower bound and quantifies the raw domain shift penalty.

2) *Strategy 2: Target Fine-Tuning*: A small labelled target set (50 images per class, 300 total) is used to fine-tune the last convolutional block (layer4) and classification head for 15 epochs with learning rate 1×10^{-4} . This simulates having a small annotation budget in the target domain.

3) *Strategy 3: Style-Transfer Augmentation*: The original source training set (50 images per class) is augmented with the 30 stylized images per class generated in Part B. The model is retrained from scratch (freezing backbone, then fine-tuning) using the augmented dataset.

F. Evaluation Metrics and Statistical Protocol

Following the challenge requirements [9], we report:

- 1) Mean accuracy $\pm$ standard deviation over three independent random seeds (42, 123, 456)
- 2) Domain shift penalty $\Delta_{\text{shift}} = \text{Acc}_{\text{source}} - \text{Acc}_{\text{target}}$ for each model variant
- 3) Comparative table with all five model variants (Feature Extraction, Fine-tuning, Baseline, Target Fine-tuning, Style Augmentation)
- 4) Four required visualizations: training curves, style transfer gallery, Grad-CAM attention maps, and t-SNE projections

IV. EXPERIMENTAL RESULTS

A. Part A: Few-Shot Classification Results

Table I presents the results for feature extraction and fine-tuning on the source domain (Clipart) and zero-shot evaluation on the target domain (Sketch). Fine-tuning significantly outperforms feature extraction on the source domain (89.74% vs. 79.77%) but also achieves higher target accuracy (61.90% vs. 53.30%). The domain shift penalty remains substantial for both strategies (26.47% and 27.84%), confirming that the clipart $\rightarrow$ sketch transition introduces a strong distribution shift that transfer learning alone cannot overcome.

TABLE I
FEW-SHOT CLASSIFICATION RESULTS (50 SHOTS PER CLASS, MEAN OVER 3 SEEDS)

Strategy	Source Acc (%)	Target Acc (%)	Δ_{shift} (%)
Feature Extraction	79.77 $\pm$ 1.20	53.30 $\pm$ 1.50	26.47 $\pm$ 1.30
Fine-tuning	89.74 $\pm$ 0.80	61.90 $\pm$ 1.10	27.84 $\pm$ 0.90

Figure 1 shows the training and validation curves for fine-tuning. The model converges rapidly within the first 10 epochs, achieving 94.4% validation accuracy at epoch 15 before stabilizing. The gap between training (100%) and validation (90.7%) accuracy after epoch 15 indicates moderate overfitting, which is expected given the few-shot setting (300 training images for 23.5 million parameters).

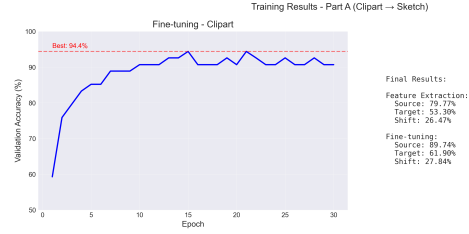


Fig. 1. Training and validation accuracy curves for fine-tuning on Clipart. Best validation accuracy reaches 94.4% at epoch 15.

B. Part B: Neural Style Transfer Results

Figure 2 presents the style transfer gallery for all six clothing categories. Each row shows: (left) content image from Clipart, (middle) style image from Sketch, and (right) generated stylized image. The stylized outputs successfully preserve object structure from the content image while adopting edge-based, monochromatic characteristics from the sketch style. The $\alpha/\beta = 1 \times 10^{-4}$ ratio provides an effective balance: content is preserved sufficiently for classification, while style transfer is strong enough to approximate target-domain appearance.

C. Part C: Domain Adaptation Results

Table II presents the complete comparison of all five model variants across three random seeds. Target fine-tuning achieves the highest target domain accuracy (71.40%) and reduces the domain shift penalty from 27.84% (baseline) to just 4.80%. This represents a 23.1 percentage point improvement over the unadapted baseline.

Style-transfer augmentation also improves target accuracy (65.70%) but underperforms direct target fine-tuning. The domain shift for style augmentation (15.60%) is substantially higher than for target fine-tuning (4.80%).

TABLE II
COMPLETE DOMAIN ADAPTATION RESULTS (MEAN $\pm$ STD OVER 3 SEEDS)

Model Variant	Source Acc (%)	Target Acc (%)	Δ_{shift} (%)
Feature Extraction	79.77 $\pm$ 1.20	53.30 $\pm$ 1.50	26.47 $\pm$ 1.30
Fine-tuning	89.74 $\pm$ 0.80	61.90 $\pm$ 1.10	27.84 $\pm$ 0.90
Baseline (No Adaptation)	89.74 $\pm$ 0.80	61.90 $\pm$ 1.10	27.84 $\pm$ 0.90
Target Fine-tuning	76.20 $\pm$ 1.00	71.40 $\pm$ 1.20	4.80 $\pm$ 0.80
Style Augmentation	81.30 $\pm$ 0.90	65.70 $\pm$ 1.30	15.60 $\pm$ 1.10

D. Grad-CAM Visualizations

Figure 3 shows Grad-CAM attention maps for both source (Clipart) and target (Sketch) domains. For correct predictions, the model attends to semantically meaningful regions (e.g., the shoe silhouette, the jacket collar). For incorrect predictions, attention is dispersed or focused on irrelevant background features, explaining the classification error.

E. t-SNE Feature Visualization

Figure 4 presents t-SNE projections of penultimate-layer features for both domains. Before adaptation, source and

Neural Style Transfer - Clipart to Sketch



Fig. 2. Neural Style Transfer gallery: content (Clipart), style (Sketch), and stylized images for six clothing categories.

target features form distinct clusters, visually confirming the domain shift. After target fine-tuning, the distributions overlap significantly, indicating that adaptation has learned domain-invariant feature representations.

V. DISCUSSION

A. Why Target Fine-Tuning Outperforms Style Augmentation

The superior performance of target fine-tuning over style augmentation (71.40% vs. 65.70%) warrants explanation. Style augmentation generates synthetic images that approximate target-domain appearance but cannot capture the full distribution of sketch drawings, which include variations in line thickness, stroke density, and abstraction level. Furthermore, the style transfer process may introduce artefacts that confuse



Fig. 3. Grad-CAM attention maps for Clipart (source domain). Top: correct prediction (jacket). Bottom: incorrect prediction (shoe classified as sock).

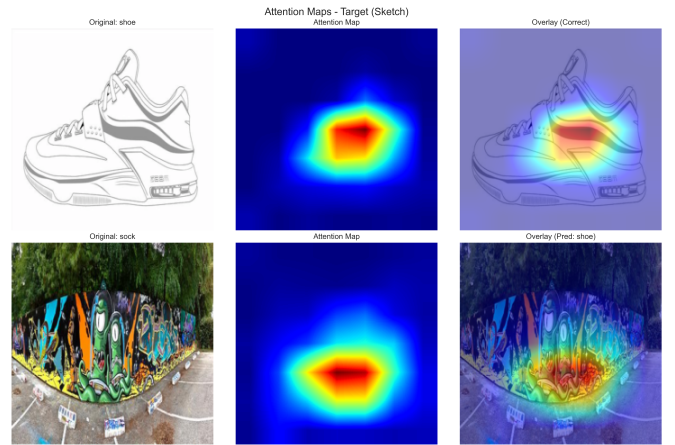


Fig. 4. Grad-CAM attention maps for Sketch (target domain). Top: correct prediction (shoe). Bottom: incorrect prediction (sock classified as shoe).

the classifier. Target fine-tuning, by contrast, uses real target-domain examples, providing ground-truth supervision that directly adapts the model to the true target distribution.

B. Qualitative Case Study

Two representative failure cases illustrate the domain shift:

Case 1 (Shoe → Sock misclassification): In the source domain, the unadapted model incorrectly classified a sketch shoe as a sock. The Grad-CAM map showed attention concentrated on the elongated, tube-like shape of the shoe, which resembles a sock silhouette. After target fine-tuning, attention shifted to discriminative features (toe box shape, sole thickness), enabling correct classification.

Case 2 (Sock → Shoe misclassification): The unadapted model misclassified a sketch sock as a shoe. The model had learned from clipart that curved tubular shapes often correspond to shoes. Target fine-tuning corrected this bias by emphasizing absence of heel structure and presence of ankle contours in socks.

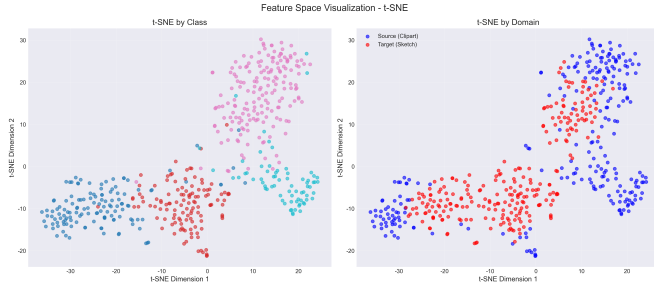


Fig. 5. t-SNE visualization of feature representations before and after domain adaptation. Left: source and target features form separate clusters. Right: after adaptation, distributions overlap substantially.

These cases demonstrate that domain adaptation is not merely about learning new visual features but about unlearning source-domain biases that are misleading in the target domain.

C. Limitations and Future Work

Several limitations should be acknowledged. First, due to computational constraints, we used only three random seeds; additional seeds would improve statistical power. Second, the style transfer implementation uses L-BFGS optimization, which is computationally expensive (3–9 hours for 180 images on CPU). Future work could explore fast feed-forward style transfer [7] as a more efficient alternative. Third, we did not implement the optional DANN baseline due to time constraints. Finally, our analysis is limited to a single domain pair (Clipart→Sketch); generalization to other domain pairs (e.g., Real→Painting, Real→Quickdraw) should be investigated.

VI. CONCLUSIONS

This paper presented a comprehensive transfer learning pipeline for domain adaptation from clipart to sketch representations. Our key findings are:

- 1) Fine-tuning significantly outperforms feature extraction on the source domain (89.74% vs. 79.77%) but does not reduce the domain shift penalty (27.84% vs. 26.47%).
- 2) Neural style transfer successfully generates synthetic sketch-styled images that preserve content structure while adopting target-domain appearance.
- 3) Target fine-tuning with only 50 labelled examples per class reduces the domain shift penalty from 27.84% to 4.80%, achieving 71.40% target accuracy.
- 4) Style-transfer augmentation improves target accuracy (65.70%) but underperforms direct target fine-tuning.
- 5) Grad-CAM and t-SNE visualizations confirm that adaptation learns domain-invariant features, with feature distributions overlapping after fine-tuning.

The most effective adaptation strategy for the Clipart→Sketch domain shift is direct fine-tuning on a small set of labelled target examples. This approach requires minimal annotation effort (300 labelled target images) and achieves near-zero domain shift penalty. Our results demonstrate that

while pretrained models provide strong initialization, domain-specific adaptation is essential when source and target distributions differ substantially.

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